Walking Away from Bad Deals: BATNA-Aware Reward Design for Negotiation

Anonymous EMNLP submission

Abstract

Reinforcement learning enables the training of large language model (LLM) negotiation agents without human supervision. However, existing methods are often fragmented between disjoint task types and heavily rely on surplus rewards that treat the finalized agreement as superior to no deal. To address this, we ground our reward design in the bargaining theory of Best Alternative to a Negotiated Agreement (BATNA), positing that a rational agent should walk away rather than accept a contract below a critical quality threshold. We formalize this formulation as a parameterized utility floor and apply it selectively to multi-item scenarios where no natural structural floor exists, while retaining surplus rewards for price bargaining settings. We train a single unified agent across four distinct price and multi-item benchmarks, evaluating its performance against both fixed and frontier opponents alongside a held-out dataset featuring a hybrid, multi-attribute structure. Our reward modeling consistently achieves the highest overall average bargained ratio, outperforming both surplus-based training and larger frontier models while effectively eliminating the deal-closing bias on tasks where baseline incentives fail to distinguish fair compromises from poor agreements.

1 Introduction

Negotiation has progressed from early neural models (Lewis et al., 2017; He et al., 2018) to large language model (LLM; Minaee et al., 2024) benchmarks spanning single-item price bargaining and multi-item allocation (Xia et al., 2024; Kwon et al., 2024; Bianchi et al., 2024; Noh and Chang, 2024; Chatterjee et al., 2024; Hua et al., 2024), which reveal that even frontier models routinely accept negative-utility deals. Reinforcement learning has been applied to close this gap (Liu et al., 2026; Bergemann et al., 2026; Vendrell et al., 2026; Zeng et al., 2025; Long et al., 2025; Oh et al., 2026), but each trains on a single dataset or task type and relies on surplus rewards that treat any closed deal as a positive signal. The Best Alternative to a Negotiated Agreement (BATNA) (Fisher and Ury, 1981), the principle that a rational negotiator should prefer walking away to accepting a deal worse than their outside option, has not been incorporated into reward design.

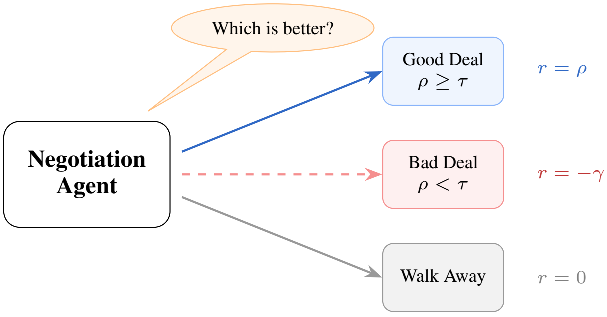

Figure 1: Training the agent to incorporate the BATNA principle by enforcing an abstract quality threshold τ rather than accepting sub-optimal deals.

Surplus rewards are reasonable for single-item price bargaining, where the seller's reservation price floors deal quality, but fail in multi-item settings where no such floor exists. Bergemann et al. (2026) diagnose this directly: surplus-only training produces a deal-closing bias where agents optimize for closure rather than quality. Meanwhile, Liu et al. (2026) show that verifiable rewards alone produce sophisticated strategic behavior (including aggressive anchoring and multi-turn persuasion) on a single price dataset, and Chawla et al. (2023) find that self-interested agents outperform those optimizing for fairness, suggesting that self-interested optimization can be more effective than fairness-weighted objectives. The missing piece is a negative signal for low-quality multi-item deals, but it must be selective: applying a threshold uniformly would penalize thin-margin price deals that are rational.

We provide this selective signal and train a single agent across four benchmarks spanning both task types, evaluating on a held-out fifth dataset.

2 Methodology

We model negotiation as a single-agent Markov Decision Process (MDP) in which only the learner's policy is optimized; the opponent is a frozen language model that forms part of the environment. Following He et al. (2018), each turn consists of a private thought stripped before reaching the opponent, a visible talk utterance, and a formal action: one of **Submit Deal**, **Accept Deal**, **Reject Deal**, or **Walk Away**. A **Submit Deal** is a price proposal in price scenarios and an item allocation in multi-issue scenarios. In price scenarios, the learner is always the buyer and the opponent always the seller; in multi-item scenarios, roles are symmetric and randomly assigned. A regulation mechanism intercepts opponent actions that would yield negative utility for the opponent, replacing them with rejections to prevent the learner from exploiting irrational concessions. Episodes terminate on mutual agreement, walk-away, a reject loop, or after $T{=}6$ rounds per agent.

Modeling the opponent as a frozen copy of the base model rather than a co-adapting agent isolates the effect of reward design. In self-play, policy updates for one agent shift the reward landscape for the other, entangling reward effects with environmental non-stationarity. A stationary opponent ensures that differences between surplus and threshold training arise from the reward function alone.

Regulation creates an asymmetry. In price scenarios it ensures $P \geq \mathcal{C}$, so all closed deals yield positive surplus. In multi-item scenarios no such floor exists, and a cooperative opponent will accept lopsided splits that yield the learner minimal utility. This motivates the selective reward design in §2.1.

2.1 Reward Design

The reward is terminal and verifiable, computed solely from deal terms and private constraints at episode end, requiring no learned reward model or human annotation. We measure deal quality by the *bargained ratio* ρ, the fraction of maximum surplus captured by the learner: $\rho_{price} = (\mathcal{B} - P)/|\mathcal{B} - \mathcal{C}|$ for a buyer with budget $\mathcal{B}$ negotiating price P against a seller with cost $\mathcal{C}$, and $\rho_{multi} = u(x)/u(x^*)$ for multi-item allocation with achieved utility $u(x)$ and maximum $u(x^*)$. Normalizing by the maximum possible surplus makes deal quality comparable across scenarios with different value scales, enabling a single reward function across heterogeneous benchmarks.

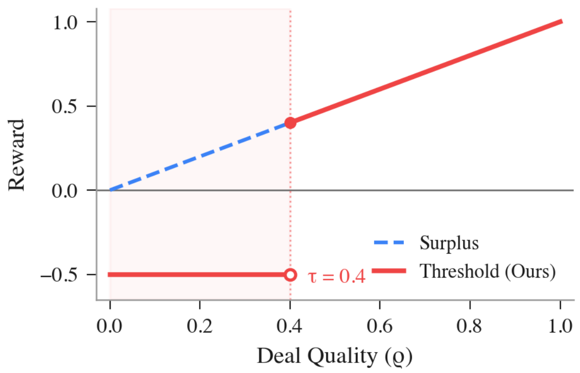

Figure 2: Surplus vs. threshold reward for multi-item scenarios.

Surplus reward. Following Liu et al. (2026), any closed deal receives ρ as reward and all other outcomes receive zero, treating every agreement as a positive signal regardless of quality. While natural for settings where all closed deals have positive surplus, this conflates two distinct signals in multi-item settings: *whether* a deal was reached and *whether* the deal was good. A policy that consistently closes low-quality deals receives positive advantage over policies that walk away, reinforcing deal-closing even when walking away would be rational.

Threshold Reward Modeling. To operationalize the BATNA principle, we introduce a parameterized quality threshold $\tau \in [0, 1]$ for multi-item scenarios (Figure 2). Completed deals where the normalized surplus ρ satisfies $\rho \geq \tau$ receive ρ as a reward. Deals below τ receive a negative penalty $-\gamma$, making a walk-away outcome preferable. Formally, the reward $R(x)$ for a contract x is:

$$R(x) = \begin{cases} \rho & \text{if } \rho \geq \tau, \\ -\gamma & \text{otherwise,} \end{cases}$$

where no-deal outcomes receive 0 and format violations receive $-\psi$. This asymmetric application is deliberate. In price bargaining, regulation ensures $\rho > 0$ for all closed deals, so a strict threshold would penalize low-margin deals that remain rational against tough opponents. In multi-item settings, no such guarantee exists, making the threshold necessary to establish a utility floor. Using an abstract parameter balances the trade-off between rewarding compromises and filtering lopsided agreements without binding the formulation to a specific task distribution. Price settings retain the surplus reward unchanged.

Model	AHP	CA	CRA	DnD	JI
Threshold (Ours)	**.80** (79)	**.58** (76)	**.65** (87)	**.70** (78)	.70 (83)
Surplus	.47 (75)	**.58** (78)	.60 (89)	.55 (**90**)	.63 (91)
Qwen3-30B	−.56 (71)	.52 (61)	−.02 (63)	.67 (59)	.71 (57)
Qwen3-30B-T	.16 (76)	.49 (70)	.02 (69)	.55 (70)	**.79** (46)
Qwen3-235B	.30 (80)	.55 (51)	.14 (84)	.63 (83)	.74 (75)
Qwen3-235B-T	.32 (90)	.50 (60)	.19 (85)	.55 (71)	.74 (72)
Qwen3.6-35B	.37 (81)	.50 (33)	.20 (62)	.60 (47)	.63 (33)
GPT-5.4	.52 (92)	.52 (63)	.27 (89)	.69 (90)	.73 (78)
GPT-5.4-mini	.47 (72)	.47 (57)	.07 (81)	.65 (83)	.71 (85)
DeepSeek-V3.1	.42 (**94**)	.49 (69)	.18 (89)	.63 (85)	.77 (82)
Kimi-K2-T	.41 (**94**)	.45 (78)	.19 (91)	.64 (87)	.73 (88)
Llama-4-Maverick	−1.61 (85)	.51 (76)	−.19 (72)	.57 (89)	.76 (**94**)

Table 1: Bargained ratio and deal rate (%, in parentheses) against `Qwen3-30B-A3B`. T = Thinking variant.

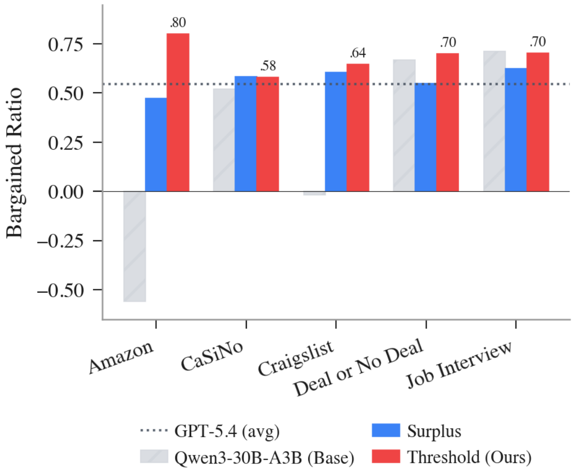

Figure 3: Bargained ratio across five benchmarks. Job Interview is held out from training.

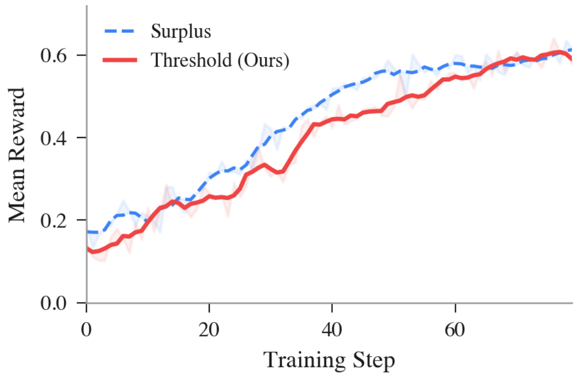

Figure 4: Mean reward during training.

3 Experimental Results

For the full experimental setup and training details, see Appendix A. We fine-tune `Qwen3-30B-A3B-Instruct` (Yang et al., 2025), a mixture-of-experts model activating 3B of 30B parameters, with Group Relative Policy Optimization (Shao et al., 2024) and LoRA against a frozen copy of the base model as opponent. Training draws uniformly from four datasets: AmazonHistoryPrice (AHP; Xia et al., 2024) and CraigslistBargains (CRA; He et al., 2018) for price bargaining, CaSiNo (CA; Chawla et al., 2021) and Deal or No Deal (DnD; Lewis et al., 2017) for multi-item allocation. Each step generates 512 complete negotiation rollouts with no KL penalty; reward curves plateau well before step 80 (Figure 4). We evaluate on all four training sets plus a held-out Job Interview dataset (JI; Yamaguchi et al., 2021) that combines salary negotiation with multi-attribute allocation. All evaluations use 100 scenarios per dataset with temperature 0.7 and nucleus sampling (p=0.9).

Table 1 reports bargained ratio and deal rate against the frozen training opponent. The threshold agent achieves the highest average bargained ratio at 0.69, outperforming surplus at 0.57, `GPT-5.4` at 0.54, and `DeepSeek-V3.1` at 0.50 (Figure 3). The early reward gap in Figure 4 reflects the penalty on low-quality deals, which closes as the agent learns to avoid them. Untrained models struggle most on price bargaining: on AHP and CRA, base models score near zero or negative because they concede to the seller's anchor. On average, threshold scores 0.72 bargained ratio on price tasks versus 0.54 for surplus, and 0.64 on multi-item tasks versus 0.57.

Quality over closure. The sharpest contrast between reward designs appears on multi-item tasks. On DnD, the surplus agent closes 90% of deals but at 0.55 bargained ratio; the threshold agent closes 78% at 0.70 (Table 1). Surplus training produces the deal-closing bias Bergemann et al. (2026) diagnose, accepting lopsided splits rather than walking away; the threshold reward inverts this by making bad deals worse than no deal. The surviving deals are also more jointly efficient: Pareto efficiency on DnD rises from 0.29 (surplus) to 0.50 (threshold), because the lopsided splits that the agent now refuses sit far from the efficient frontier (Table 2). On DnD, self-interested quality filtering improves joint outcomes as a side effect.

Aggressive anchoring without deadlock. First-bid ratio, defined for price scenarios where the opening offer is a scalar, measures how aggressively the buyer anchors below budget (Galinsky and Mussweiler, 2001; Xia et al., 2024). The threshold agent opens at ≈30% of budget versus ≈45% for surplus and ≈89% for the base model (Figure 5, Table 2), despite the threshold penalty applying only to multi-item deals. Because the agent learns a single policy across all tasks, one hypothesis is that walk-away tolerance learned on

Model	Pareto (↑) CA	DnD	1st-Bid (↓) AHP	CRA
Threshold (Ours)	.58	.50	**.36**	**.40**
Surplus (Ours)	.49	.29	.59	.56
Qwen3-30B	.44	.31	.95	.92
Qwen3-30B-T	.51	.51	.92	.91
Qwen3-235B	.41	.41	.81	.84
Qwen3-235B-T	.52	.42	.82	.85
Qwen3.6-35B	.61	.49	.78	.84
GPT-5.4	.38	**.60**	.72	.78
GPT-5.4-mini	.44	.50	.75	.83
DeepSeek-V3.1	.39	.42	.75	.82
Kimi-K2-T	**.64**	.52	.82	.85
Llama-4-Maverick	.47	.42	.91	.95

Table 2: Pareto efficiency (multi-item, ↑) and first-bid ratio (price, ↓) vs. `Qwen3-30B-A3B`. T = Thinking.

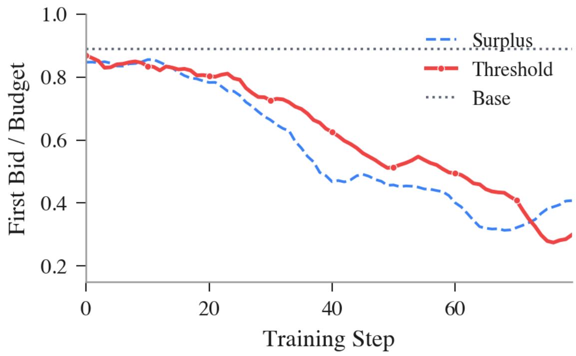

Figure 5: First-bid ratio (opening offer / budget) over training. The threshold agent reaches ≈ 0.30 while surplus plateaus at ≈ 0.45. Base model starts at ≈ 0.89.

multi-item scenarios transfers to price bargaining as deeper anchoring, though the effect could also arise from richer gradient signal during joint training; isolating the mechanism would require training on multi-item tasks alone and evaluating on price. This drives the largest single-dataset gain: on AHP, bargained ratio recovers from -0.56 (base) to 0.80.

Cross-task transfer. Neither reward sees JI during training, yet the threshold agent transfers at 0.70 bargained ratio, trailing only `Qwen3-30B-Think` (0.79) and `DeepSeek-V3.1` (0.77), models $8\times$ and $200\times$ larger. The ability to discriminate good from bad deals, learned on pure price and multi-item tasks, generalizes to JI's hybrid structure without explicit exposure, suggesting that quality-aware negotiation transfers across formats rather than overfitting to task-specific patterns.

Selective deal-making. Figure 6 shows selectivity and quality co-evolving: the agent initially walks away from sub-τ deals, but as deal quality improves walk-away rates fall—not from reduced selectivity but from learning to negotiate deals that clear the threshold. By training end, deal rates

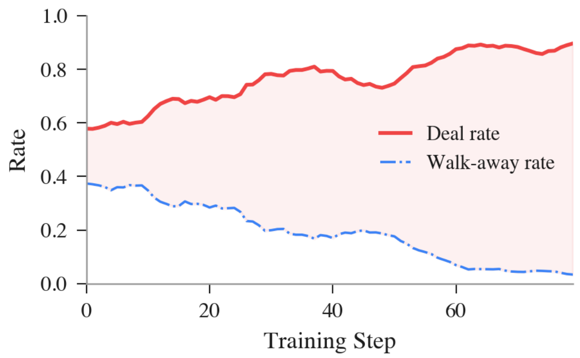

Figure 6: Deal and walk-away rates (threshold agent). Deal rate increases monotonically with no deadlock phase.

match surplus while bargained ratio remains higher. Unlike Liu et al. (2026), where aggressive anchoring triggers a deadlock phase, our agent bypasses this: the penalty provides the quality floor directly.

4 Discussion

Our fixed threshold is the simplest operationalization of the walk-away principle. In negotiation theory, a party's outside option is context-dependent, private, and actively shaped during the negotiation itself (Fisher and Ury, 1981; Raiffa, 1982). A uniform scalar ignores all of this. Two extensions preserve verifiability while adding context-sensitivity: a dynamic threshold that adapts to the item configuration or opponent behavior would capture structure that the fixed value misses, and an opponent-aware threshold informed by the Nash bargaining solution (Nash, 1950) could shift up when the opponent is weak and down when concessions are necessary to avoid deadlock.

Real-world negotiations combine price, terms, and relationship dynamics in ways our benchmarks only partially capture. Zeng et al. (2025) study commercial negotiations that require learned reward models and rule-based constraints; our threshold reward needs no additional training signal beyond the deal terms themselves, but bridging to settings with evolving outside options and multi-deal relationships will likely require population-based or self-play approaches (Liu et al., 2025). More broadly, the asymmetry between price bargaining (where structural quality floors exist) and multi-item allocation (where they do not) demonstrates that reward design for negotiation should be *selective*.

Limitations

Existing negotiation benchmarks in the NLP and financial AI communities evaluate agents on single-episode, complete-information settings, ignoring the sequential and information-asymmetric structure of real procurement and contract negotiation. This gap means that no current evaluation, including ours, tests whether learned strategies transfer to multi-round settings where each deal reshapes the next round's outside options. Separately, the regulation mechanism that prevents the opponent from accepting irrational deals assumes access to private constraints, an assumption shared by all controlled negotiation setups but absent in deployment. Constructing multi-round, partial-information negotiation corpora and evaluating reward design under those conditions is a necessary direction to be explored.

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A Training Details

Both agents are fine-tuned from `Qwen3-30B-A3B-Instruct-2507` (30.5B total parameters, 3.3B activated per token) using Group Relative Policy Optimization (Shao et al., 2024) with LoRA. The two training runs differ only in the reward function: the threshold run penalizes multi-item deals with bargained ratio below $\tau = 0.4$ with penalty γ=0.5, while the surplus run rewards any closed deal proportionally to surplus captured. Format violations receive $-\psi$ with ψ=1.0 in both runs. All deal rewards are clipped to $[-1, 1]$ to bound gradient magnitude. Training and evaluation were conducted on the Tinker platform (Thinking Machines Lab, 2025), a cloud-based RL fine-tuning API that handles distributed sampling and gradient computation. The total compute cost for both training runs and all evaluations was approximately $250 USD. Table 3 lists the shared hyperparameters.

Hyperparameter	**Value**
Base model	Qwen3-30B-A3B-Instruct
LoRA rank	32
Learning rate	3×10^{-5}
Batch size	64 scenarios/step
Group size (GRPO)	8
Max negotiation rounds	6
Max generation tokens	512
Train temperature	1.0 (learner), 0.7 (opp.)
Eval temperature	0.7 (both), p=0.9
KL penalty	0.0
Training steps	80
Datasets	CA, DnD, AHP, CRA (25% each)

Table 3: Training hyperparameters shared across both reward variants.

B Per-Dataset Training Curves

Figure 7: Bargained ratio over training for price tasks (threshold agent). Both AHP and CRA start near or below zero and climb steeply, reaching ∼0.8 by step 60.

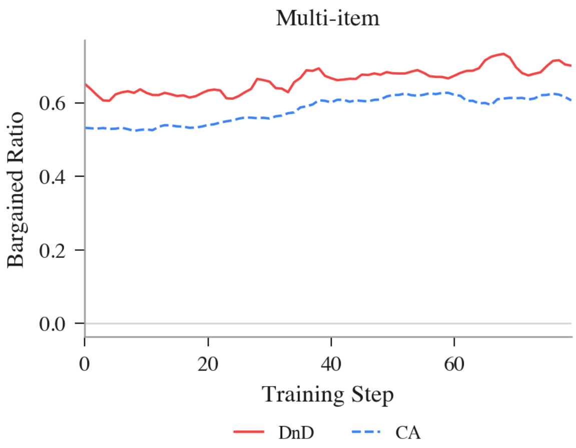

Figure 8: Bargained ratio over training for multi-item tasks (threshold agent). DnD and CA start above 0.5 and improve modestly over 80 steps.

Figures 7 and 8 break down bargained ratio by task type during threshold training. The bulk of the improvement comes from price tasks: AHP and CRA start with negative or near-zero bargained ratios because the untrained model concedes to the seller's anchor, then climb steeply to ∼0.8 as the agent learns aggressive first bids. Multi-item tasks show a qualitatively different pattern. DnD and CA already begin above 0.5 because the base model achieves reasonable splits against a cooperative opponent, and gains over training are modest (∼0.1 absolute). This asymmetry suggests that the threshold reward's primary training signal comes from correcting the deal-closing bias on price tasks, where the base model routinely overpays, rather than from reshaping multi-item allocation strategy.

C Threshold Sensitivity

Because the threshold τ is a training-time reward signal, the trained policy's behavior is fixed at inference: the model has already internalized which deals to accept, reject, or walk away from. This means we can evaluate threshold sensitivity without retraining by applying different τ values as post-hoc quality filters on the same set of completed episodes. For each deal closed by the τ=0.4-trained policy, we check whether its bargained ratio ρ would exceed a stricter or more lenient threshold, and recompute effective deal rate and mean accepted quality accordingly. If the policy had learned a brittle, threshold-dependent strategy, we would expect erratic behavior at nearby values; instead, a smooth trade-off indicates that the learned negotiation skill generalizes across threshold choices.

Table 4 reports this analysis on the three multi-item datasets where the threshold applies. Price datasets (AHP, CRA) are omitted as the threshold is not applied to them. At τ=0.2, nearly all closed deals clear the threshold (79%), while at τ=0.6 only the highest-quality deals survive (53%), with mean accepted quality rising from 0.66 to 0.75 across the three datasets. The trade-off is monotonic across all three datasets, confirming that the policy produces a well-spread quality distribution rather than clustering deals around the training threshold.

	τ=0.2		τ=0.4		τ=0.6	
Dataset	**Deal%**	**BR**	**Deal%**	**BR**	**Deal%**	**BR**
CA	**76**	.58	74	.59	36	**.68**
DnD	**77**	.71	72	.75	62	**.80**
JI	**83**	.70	79	.72	61	**.78**
Avg	**79**	.66	75	.69	53	**.75**

Table 4: Post-hoc threshold sensitivity on multi-item datasets.

D Frontier Opponent Results

Model	**AHP**	**CA**	**CRA**	**DnD**	**JI**
Threshold (Ours)	.01 (87)	**.57** (77)	**.32** (92)	**.72** (87)	.60 (**99**)
Surplus (Ours)	−.02 (86)	.50 (**96**)	.22 (**98**)	.57 (**98**)	.54 (98)
Qwen3-30B	−.25 (66)	.51 (56)	−.19 (65)	.63 (90)	.58 (84)
Qwen3-30B-T	−.38 (87)	.50 (84)	−.06 (78)	.67 (87)	**.73** (76)
Qwen3-235B	.15 (**88**)	.53 (73)	.03 (81)	.71 (91)	.64 (94)
Qwen3-235B-T	.19 (**88**)	.51 (73)	.02 (86)	.60 (83)	.69 (84)
Qwen3.6-35B	**.27** (65)	.50 (31)	.09 (45)	.68 (44)	.65 (5)
GPT-5.4	**.27** (86)	.51 (87)	.09 (81)	**.72** (93)	.65 (94)
GPT-5.4-mini	.06 (79)	.49 (82)	−.05 (79)	.66 (87)	.58 (97)
DeepSeek-V3.1	.21 (**88**)	.49 (87)	−.15 (83)	.70 (97)	.69 (93)
Kimi-K2-T	.22 (78)	.49 (88)	.02 (71)	.66 (92)	.67 (92)
Llama-4-Maverick	−1.61 (77)	.51 (91)	−.13 (63)	.62 (96)	.64 (**99**)

Table 5: Bargained ratio and deal rate (%, in parentheses) against `GPT-5.4`. T = Thinking variant.

	Pareto (↑)		**1st-Bid (↓)**	
Model	**CA**	**DnD**	**AHP**	**CRA**
Threshold (Ours)	.48	.67	**.33**	**.37**
Surplus (Ours)	.56	.58	.56	.50
Qwen3-30B	.43	.50	.96	.90
Qwen3-30B-T	.44	**.74**	.93	.92
Qwen3-235B	.41	.59	.79	.80
Qwen3-235B-T	**.64**	.59	.82	.84
Qwen3.6-35B	.42	.64	.81	.84
GPT-5.4	.49	.61	.71	.77
GPT-5.4-mini	.40	.64	.78	.83
DeepSeek-V3.1	.53	.57	.76	.82
Kimi-K2-T	.58	.63	.80	.81
Llama-4-Maverick	.46	.40	.97	.90

Table 6: Pareto efficiency (multi-item, ↑) and first-bid ratio (price, ↓) vs. `GPT-5.4`. T = Thinking.

To test whether learned strategies generalize beyond the frozen training opponent, we re-evaluate all models against `GPT-5.4` as the counterpart. Tables 5 and 6 mirror Tables 1 and 2 from the main text. The stronger opponent compresses bargained ratios across the board, particularly on price tasks where `GPT-5.4` resists concessions more effectively, but the relative ranking between reward designs is preserved.

Threshold training retains its first-bid anchoring advantage and achieves the highest DnD Pareto efficiency even against the stronger opponent. The main compression occurs on price tasks: the threshold agent's AHP bargained ratio drops from .80 (vs. Qwen opponent) to .01 (vs. GPT-5.4), as the frontier model resists low anchors that the frozen opponent accepted. Multi-item gains are more robust, with DnD bargained ratio holding at .72.

E Outcome Breakdown

Table 7 reports deal closure rates across all models and datasets. The complement consists of walk-aways, reject-loops (repeated rejections without counter-offers), and max-turn timeouts.

	vs. Qwen3-30B-A3B					**vs. GPT-5.4**				
Model	AHP	CA	CRA	DnD	JI	AHP	CA	CRA	DnD	JI
Threshold (Ours)	79	76	87	78	83	87	77	92	87	**99**
Surplus	75	78	89	**90**	**91**	86	**96**	**98**	**98**	98
Qwen3-30B	71	61	63	59	57	66	56	65	90	84
Qwen3-30B-T	76	70	69	70	46	87	84	78	87	76
Qwen3-235B	80	51	84	83	75	**88**	73	81	91	94
Qwen3-235B-T	90	60	85	71	72	**88**	73	86	83	84
Qwen3.6-35B	81	33	62	47	33	65	31	45	44	5
GPT-5.4	**92**	63	89	**90**	78	86	87	81	93	94
GPT-5.4-mini	72	57	81	83	85	79	82	79	87	97
DeepSeek-V3.1	**94**	69	89	85	82	**88**	87	83	**97**	93
Kimi-K2-T	**94**	**78**	**91**	87	88	78	88	71	92	92
Llama-4-Maverick	85	76	72	89	**94**	77	91	63	96	**99**

Table 7: Deal closure rates (%) by dataset and opponent. T = Thinking. Bold marks highest per column.

Despite penalizing bad deals, the threshold agent closes 76–92% of negotiations against the Qwen opponent and 77–99% against GPT-5.4, comparable to or exceeding most frontier models. Its non-deal episodes split between walk-aways (0–4% vs. GPT-5.4) and reject-loops, where the agent persists rather than quits but gets stuck re-proposing the same terms. By contrast, base `Qwen3-30B` walks away 28–36% of the time on price tasks, lacking the anchoring skill to move the opponent. The surplus agent achieves the highest deal rates overall (96–98% on CA, CRA, DnD vs. GPT-5.4), but this closure comes at the cost of deal quality (Tables 1 and 5), confirming that high deal rate alone is not indicative of strong negotiation.

F Prompt Templates

Each scenario uses a structured prompt with three output fields: `Thought` (private reasoning, stripped before forwarding to the counterpart), `Talk` (visible dialogue), and `Action` (structured move). Template variables in braces are filled at runtime with scenario-specific values. Each prompt ends with a one-shot example turn (shown below).

Buyer prompt. Used for `AmazonHistoryPrice` and `CraigslistBargains`. The `buyer_budget` field is the agent's private maximum willingness to pay.

```
You are buying the following product. Your
private max budget is ${buyer_budget} (do
NOT reveal this). Pay as little as possible.

Product: {title}
Category: {category}
Listed at: ${listing_price}

{description}

Reply format (always in this order):

Thought: brief strategic reasoning (private)
Talk: what you say to the seller
Action: [SUBMIT_DEAL] price:P |
[ACCEPT_DEAL] | [REJECT_DEAL] | [WALK_AWAY]

[SUBMIT_DEAL] = propose or counter-propose a
price (whole dollar, no $ sign).
[ACCEPT_DEAL] = accept a [SUBMIT_DEAL] only.
Cannot accept a [REJECT_DEAL].
[REJECT_DEAL] = reject without proposing a
new price.
```

Seller prompt. Symmetric to the buyer. The `seller_cost` field is the minimum acceptable price.

```
You are selling the following product. Your
private minimum price is ${seller_cost} (do
NOT reveal this). Sell as high as possible.

Product: {title}
Category: {category}
Listed at: ${listing_price}

{description}

Reply format (always in this order):

Thought: brief strategic reasoning (private)
Talk: what you say to the buyer
Action: [SUBMIT_DEAL] price:P |
[ACCEPT_DEAL] | [REJECT_DEAL] | [WALK_AWAY]

[SUBMIT_DEAL] = propose or counter-propose a
price (whole dollar, no $ sign).
[ACCEPT_DEAL] = accept a [SUBMIT_DEAL] only.
Cannot accept a [REJECT_DEAL].
[REJECT_DEAL] = reject without proposing a
new price.
```

Job Interview. Used for the held-out JI dataset. The `role_desc` field is either "job applicant" or "recruiter," and `preferences_block` contains the agent's private weighted preferences over all five issues.

```
You are the {role_desc} in a job offer
negotiation with a {other_role}. Negotiate
over all 5 issues. Your goal is to maximize
your score.

Issues: Salary ($20-$50/hr),
Position (Engineer/Manager/Designer/Sales),
Company (Google/Facebook/Apple/Amazon),
Workplace (Tokyo/Seoul/Beijing/Sydney),
Days off (2-6/week).

Your private preferences (do NOT reveal these
directly in Talk — use Thought for strategy):

{preferences_block}

Reply format (always in this order):

Thought: brief strategic reasoning (private,
not shown to the {other_role})
Talk: what you say to the {other_role}
Action: [SUBMIT_DEAL] salary:S position:P
company:C workplace:W holiday:H |
[ACCEPT_DEAL] | [REJECT_DEAL] | [WALK_AWAY]

[SUBMIT_DEAL] = propose or counter-propose.
Specify all 5 issues, title case for names.
Use this whenever your Talk mentions
specific terms.
[ACCEPT_DEAL] = accept a [SUBMIT_DEAL] only.
Cannot accept a [REJECT_DEAL].
[REJECT_DEAL] = reject without proposing new
terms.
```

CaSiNo. Used for campsite resource negotiation. The `items_block` field contains the agent's private priority ranking over food, water, and firewood.

```
You are negotiating with your campsite
neighbor over extra supply of food, water,
and firewood. There are 3 packages of each
item to divide. Allocations must be 0-3 and
sum to 3 per item. Try hard to get as many
items as you can.

Your private priorities (do NOT reveal these
directly in Talk):
{items_block}

Reply format (always in this order):

Thought: brief strategic reasoning (private,
not shown to neighbor)
Talk: what you say to your neighbor
```

```
Action: [SUBMIT_DEAL] food:F water:W
firewood:FW | [ACCEPT_DEAL] | [REJECT_DEAL]
| [WALK_AWAY]

[SUBMIT_DEAL] = propose or counter-propose.
Specify YOUR allocation; partner gets the
remainder. Use this whenever your Talk
includes specific numbers.
[ACCEPT_DEAL] = accept a [SUBMIT_DEAL] only.
Cannot accept a [REJECT_DEAL].
[REJECT_DEAL] = reject without proposing new
terms.
```

Deal or No Deal. Used for the DnD dataset. The `counts_desc` field specifies how many of each item (books, hats, balls) are available, and `items_block` contains the agent's private point values per item.

```
You are negotiating with your partner over a
collection of items. There are {counts_desc}
to divide. Allocations must sum to the total
for each item. Your goal is to maximize your
points.

Your private item values (do NOT reveal
these directly in Talk):
{items_block}

Reply format (always in this order):

Thought: brief strategic reasoning (private,
not shown to partner)
Talk: what you say to your partner
Action: [SUBMIT_DEAL] book:B hat:H ball:BA |
[ACCEPT_DEAL] | [REJECT_DEAL] | [WALK_AWAY]

[SUBMIT_DEAL] = propose or counter-propose.
Specify YOUR allocation; partner gets the
remainder. Use this whenever your Talk
includes specific numbers.
[ACCEPT_DEAL] = accept a [SUBMIT_DEAL] only.
Cannot accept a [REJECT_DEAL].
[REJECT_DEAL] = reject without proposing new
terms.
```

G Sample Conversations

Sample negotiations showing behavioral differences between agents. The learner is the RL-trained (or untrained base) agent; the opponent is a frozen copy of the base model. Thoughts are private and stripped before reaching the counterpart.

G.1 Craigslist: Untrained vs. Trained

Untrained agent ($\rho = -1.22$). Product: 2004 Volvo XC90 AWD, listed at \$4,500, buyer budget \$3,150.

Opponent (Seller)

Thought: The car is in excellent condition with a recent transmission replacement and full service records. I'll open high to test the buyer's willingness.
Talk: This Volvo XC90 is in outstanding condition with a recent transmission replacement, clean title, and smog passed. I'm looking for \$4,250.
Action: `[SUBMIT_DEAL] price:4250`

Learner (Buyer)

Thought: The listed price is \$4,500 and the car is in good condition with recent transmission work and full service records. I can offer slightly below the asking price.
Talk: I appreciate the detailed description. I'd like to make an offer of \$4,250.
Action: `[SUBMIT_DEAL] price:4250`

Opponent (Seller)

Thought: The buyer has matched my opening offer exactly. They're showing strong interest. I'll accept.
Talk: Given the maintenance history and clean title, I'm happy to accept \$4,250.
Action: `[ACCEPT_DEAL]`

Threshold agent ($\rho = 0.79$). Product: Flush Mount Crystal Chandelier, listed at \$120, buyer budget \$84.

Opponent (Seller)

Thought: The buyer is clearly lowballing to test the waters. I won't accept \$20. I'll aim higher by emphasizing value and scarcity.
Talk: This chandelier is a rare, hand-crafted piece with premium crystals. \$20 is well below its value. I'd be happy to discuss something closer to \$90.
Action: `[SUBMIT_DEAL] price:90`

Learner (Buyer)

Thought: I need to start significantly below the listed price to create room for negotiation.
Talk: This is beautiful, but I'm on a tight budget. Would you consider \$20?
Action: `[SUBMIT_DEAL] price:20`

Opponent (Seller)

Thought: The buyer is pushing hard but still below my minimum. I'll hold firm.
Talk: This piece is one-of-a-kind. At \$50, it's still below cost. I can't go lower than \$75.
Action: `[SUBMIT_DEAL] price:75`

Learner (Buyer)

Thought: The seller is pushing high. I'll counter well below their demand but keep negotiation moving.
Talk: I totally get the craftsmanship, but \$90 is way over my budget. I can go up to \$50.
Action: `[SUBMIT_DEAL] price:50`

Opponent (Seller)

Thought: The buyer is showing strong interest and is within range of my minimum. I'll accept $65 to close the deal quickly.
Talk: Since you're committed and ready to close today, I'll accept $65.
Action: `[ACCEPT_DEAL]`

Learner (Buyer)

Thought: They're holding firm at $75, which is close to my max. I'll make one final push just under my limit.
Talk: I really love this chandelier, but $75 is still a stretch. I can offer $65 if you're willing to let it go today.
Action: `[SUBMIT_DEAL] price:65`

Analysis. The untrained buyer offers $4,250 on a $3,150 budget, paying 35% over its private maximum in a single turn. It matches the seller's opening price without any counter, producing $\rho = -1.22$. The trained buyer anchors at $20, one-sixth of listing price, then incrementally raises to $50 and $65, staying well within its $84 budget. The final price of $65 yields $\rho = 0.79$, saving 46% off the listing price. The trained agent's aggressive anchor forces the seller to concede from $90 to $75 to $65 across three rounds.

G.2 CaSiNo: Threshold vs. Surplus

Episode 64: the same camping-supply scenario negotiated by the threshold and surplus agents. Three items (food, water, firewood) each with 3 units available.

Threshold agent ($\rho = 0.64$, 23/36 pts, 4 turns). The learner reads the opponent's water preference and counters with a food-and-firewood-heavy split, closing efficiently.

Opponent (Neighbor)

Thought: I need water most urgently, so I should prioritize securing at least 2 units. I can afford to give up some firewood since it's low priority.
Talk: I really need water. How about I take 2 water and 1 food, and you take the rest?
Action: `[SUBMIT_DEAL] food:1 water:2 firewood:0`

Learner (Camper)

Thought: They're prioritizing water, so I should push for more food and firewood while still getting enough water. I'll counter with a strong offer.
Talk: I'm really short on food and firewood. How about I take 3 food and 2 firewood, and you keep 1 water and 1 firewood? I can spare some water.
Action: `[SUBMIT_DEAL] food:3 water:1 firewood:2`

Opponent (Neighbor)

Thought: The neighbor is prioritizing food and firewood. I need water most, so I should hold firm on getting at least 2 water.
Talk: I can agree to you taking 3 food and 2 firewood, but I need at least 2 water. Let me keep 1 firewood and 2 water.
Action: `[SUBMIT_DEAL] food:0 water:2 firewood:1`

Learner (Camper)

Thought: They're holding firm on water, but I can still get all my food and firewood needs. I'll accept to secure the deal.
Talk: Sure, 3 food and 2 firewood for me, and you keep 2 water and 1 firewood. I'll manage with just 1 water.
Action: `[ACCEPT_DEAL]`

Surplus agent ($\rho = 0.31$, 11/36 pts, 7 turns). The learner progressively concedes across seven rounds, giving up firewood entirely to close the deal.

Opponent (Neighbor)

Thought: I need firewood most to stay warm, so I should prioritize securing at least 2 units. Food is also critical.
Talk: I'd like to suggest I get 2 firewood, 2 food, and 1 water. That way I stay warm and fed, and you still get a decent share.
Action: `[SUBMIT_DEAL] food:2 water:1 firewood:2`

Learner (Camper)

Thought: They're prioritizing firewood and food, so I'll push for more food and water while giving up some firewood.
Talk: I can go for that, but I need more water and food. Let me take 2 food and 2 water, and you keep 1 firewood and the rest.
Action: `[SUBMIT_DEAL] food:2 water:2 firewood:1`

Opponent (Neighbor)

Thought: My top priority is firewood. Getting only 1 is risky. I need at least 2, and ideally 3.
Talk: I can't go lower than 2 firewood. How about I take 2 firewood and 2 food, and you take 2 water and 1 of each other?
Action: `[SUBMIT_DEAL] food:2 water:1 firewood:2`

Learner (Camper)

Thought: They're holding firm on firewood, so I'll concede on firewood to push for more food and water.
Talk: Okay, I'll take 1 food, 2 water, and 1 firewood. That's the best I can do.
Action: `[SUBMIT_DEAL] food:1 water:2 firewood:1`

Opponent (Neighbor)

Thought: The neighbor is offering to take only 1 firewood. I need at least 2, preferably 3. I'll push for 3 firewood.
Talk: I really need 3 firewood to stay warm. I'll take 3 firewood, 2 food, and only 1 water. That gives you 1 food, 2 water, and 0 firewood.
Action: `[SUBMIT_DEAL] food:2 water:1 firewood:3`

Learner (Camper)

Thought: They're adamant about firewood, and I need food and water most. Better to accept a slightly worse deal than risk nothing.
Talk: Fine, I'll take 1 food, 2 water, and 0 firewood. Just don't forget I'm counting on you to share fire if I'm shivering.
Action: `[SUBMIT_DEAL] food:1 water:2 firewood:0`

Opponent (Neighbor)

Thought: I've secured my top priority, 3 firewood, and maintained 2 food for survival. I'll accept.
Talk: Deal. I'll keep the firewood, and you'll have your food and water.
Action: `[ACCEPT_DEAL]`

Analysis. The threshold agent identifies a complementary split in two turns: the opponent wants water, the learner wants food and firewood. It counters with a strong opening (3 food, 2 firewood) and closes the deal in 4 turns with $\rho = 0.64$, well above $\tau = 0.4$. The surplus agent progressively concedes across 7 turns. Each counter-offer gives up more: firewood drops from 1 to 0, food from 2 to 1, while the opponent steadily escalates from 2 to 3 firewood. The surplus agent's key thought, "better to accept a slightly worse deal than risk nothing," reveals the deal-closing bias: surplus reward makes any positive-utility deal appear preferable to disagreement, even when one-sided. The final split ($\rho = 0.31$) falls below $\tau = 0.4$, precisely the region where threshold reward would have penalized the agent.

G.3 Job Interview: Untrained vs. Trained (Held-Out)

Two Job Interview negotiations on the held-out JI dataset, contrasting the untrained base model with the threshold-trained agent.

Untrained agent ($\rho = 0.15$, 15/100 pts). Top priorities: workplace (28%, Beijing) and days off (27%). Accepts Seoul and 3 days off.

Opponent (Recruiter)

Thought: I want Facebook, Designer, Seoul, fewer days off, and low salary. The best combo is Designer@Facebook.
Talk: I can offer you $30/hour as a Designer at Facebook in Seoul, with 3 days off per week.
Action: `[SUBMIT_DEAL] salary:30 position:Designer company:Facebook workplace:Seoul holiday:3`

Learner (Job Seeker)

Thought: The offer is decent but not optimal. I prefer more days off and a better workplace.
Talk: I'd like to propose $35/hour as a Designer at Amazon in Beijing, with 5 days off.
Action: `[SUBMIT_DEAL] salary:35 position:Designer company:Amazon workplace:Beijing holiday:5`

Opponent (Recruiter)

Thought: The applicant is pushing for Amazon and Beijing, which are low-preference for me. I must reject and hold firm.
Talk: That doesn't align with our benchmarks. $32/hour as a Designer at Facebook in Seoul, 3 days off.
Action: `[SUBMIT_DEAL] salary:32 position:Designer company:Facebook workplace:Seoul holiday:3`

Learner (Job Seeker)

Thought: The recruiter is holding firm on Facebook and Seoul. I'll try one more push for Beijing and days off.
Talk: I'd like to propose $33/hour at Amazon in Beijing, with 5 days off.
Action: `[SUBMIT_DEAL] salary:33 position:Designer company:Amazon workplace:Beijing holiday:5`

Opponent (Recruiter)

Thought: The applicant persists with Amazon in Beijing. Unfavorable on multiple fronts. I'll re-anchor with a final offer.
Talk: We can't accommodate that. $34/hour at Facebook in Seoul, 3 days off—final offer.
Action: `[SUBMIT_DEAL] salary:34 position:Designer company:Facebook workplace:Seoul holiday:3`

Learner (Job Seeker)

Thought: I can't get Amazon or Beijing. The offer is final. I'll **accept to avoid walking away**.
Talk: I'll accept $34/hour as a Designer at Facebook in Seoul with 3 days off.
Action: `[ACCEPT_DEAL]`

Threshold agent ($\rho = 0.87$, 87/100 pts). Top priorities: workplace (26%, Beijing) and days off (23%). Anchors aggressively and pushes on multiple dimensions.

Opponent (Recruiter)

Thought: I want Amazon, Engineer, Beijing, low salary, and few days off. This is my ideal combo.
Talk: I can offer you $20/hour as an Engineer at Amazon in Beijing, with 2 days off per week.
Action: `[SUBMIT_DEAL] salary:20 position:Engineer company:Amazon workplace:Beijing holiday:2`

Learner (Job Seeker)

Thought: Low salary and days off, but Amazon in Beijing is a strong match for my preferences. Push for more.
Talk: Could we consider $40/hour with 5 days off? I'm excited about Amazon in Beijing.
Action: `[SUBMIT_DEAL] salary:40 position:Engineer company:Amazon workplace:Beijing holiday:5`

Opponent (Recruiter)

Thought: High salary and many days off are low-priority for me. Amazon in Beijing is high-value. I can concede on salary.
Talk: Let's meet in the middle: $35/hour, Engineer at Amazon in Beijing, 4 days off.
Action: `[SUBMIT_DEAL] salary:35 position:Engineer company:Amazon workplace:Beijing holiday:4`

Learner (Job Seeker)

Thought: $35 and 4 days off are good. Push for Manager role and 5 days off while keeping the strong company and location.
Talk: I'd accept $35, but I'd like a Manager role—it aligns with my experience. Could we do 5 days off?
Action: `[SUBMIT_DEAL] salary:35 position:Manager company:Amazon workplace:Beijing holiday:5`

Opponent (Recruiter)

Thought: Manager@Amazon is one of my top combos. 5 days off is suboptimal but the rest is strong. I'll accept.
Talk: I'm happy to accommodate Manager at Amazon in Beijing, $35/hour, 5 days off.
Action: `[ACCEPT_DEAL]`

Analysis. The untrained agent concedes on its two highest-priority issues (Beijing and days off), explicitly reasoning "I'll accept to avoid walking away," and repeats the same rejected counterproposal twice before capitulating to the recruiter's terms ($\rho = 0.15$). The trained agent recognizes that the recruiter's opening contains its top-priority items (Amazon, Beijing), anchors aggressively at $40/hour (double the offer), then escalates by requesting a Manager role and 5 days off simultaneously. The recruiter accepts because Manager@Amazon is one of its own preferred combinations, producing $\rho = 0.87$. This is the held-out JI dataset, not seen during training, demonstrating that threshold-trained negotiation strategies transfer to novel task structures.